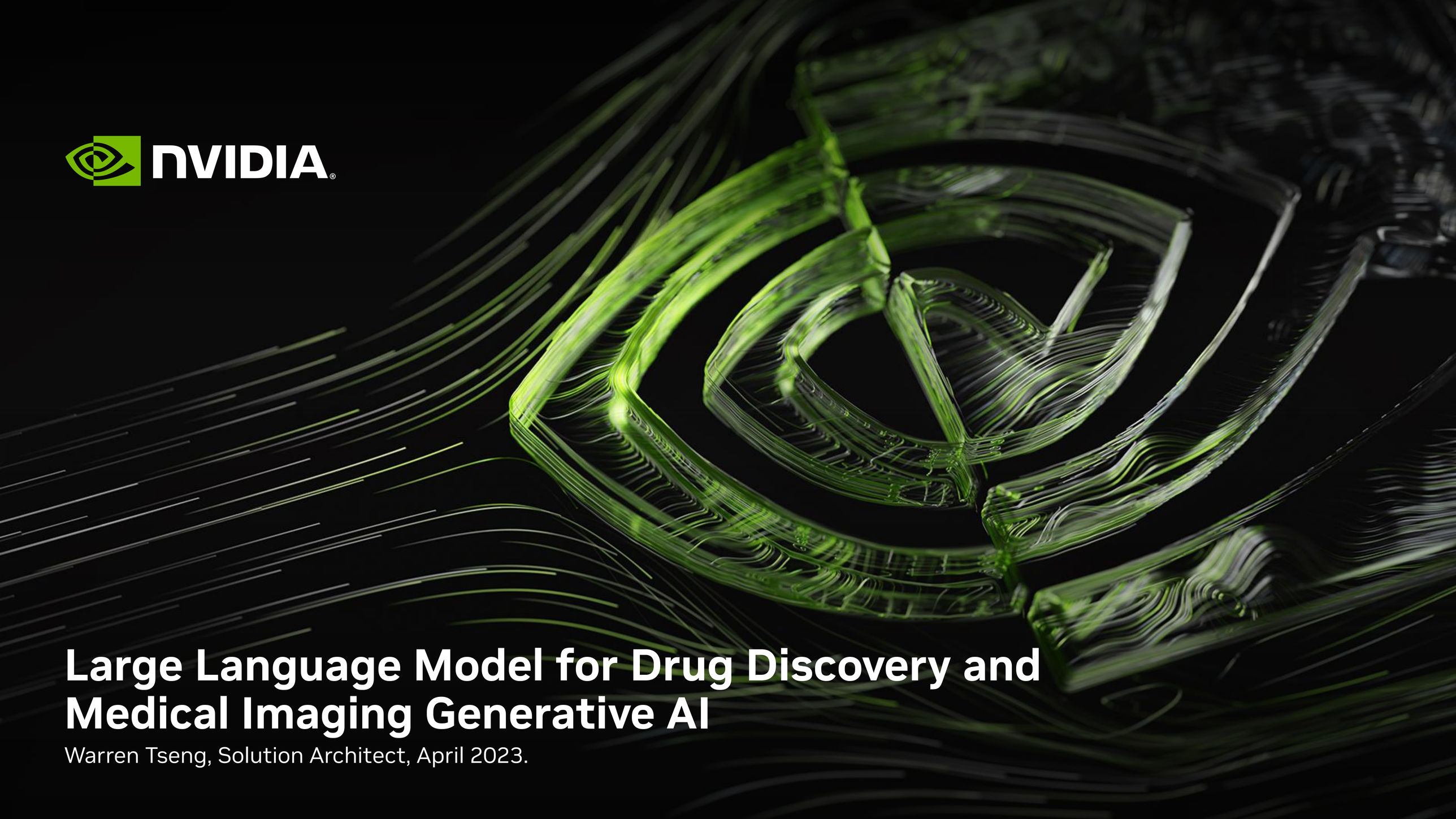


Large Language Model for Drug Discovery and Medical Imaging Generative AI

Warren Tseng, Solution Architect, April 2023.

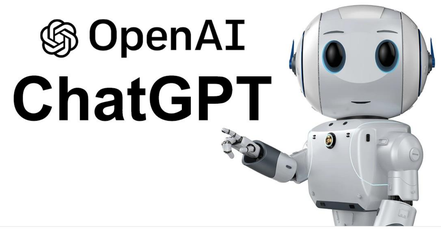


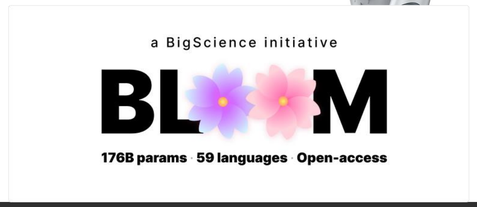
- Large Language Model (LLM) and Biology?
- Use Cases: LLM for Drug Discovery Acceleration
- NVIDIA BioNeMo
- MONAI Generative AI

Large Language Model (LLM) and Biology?



2023 Large Language Model (LLM)



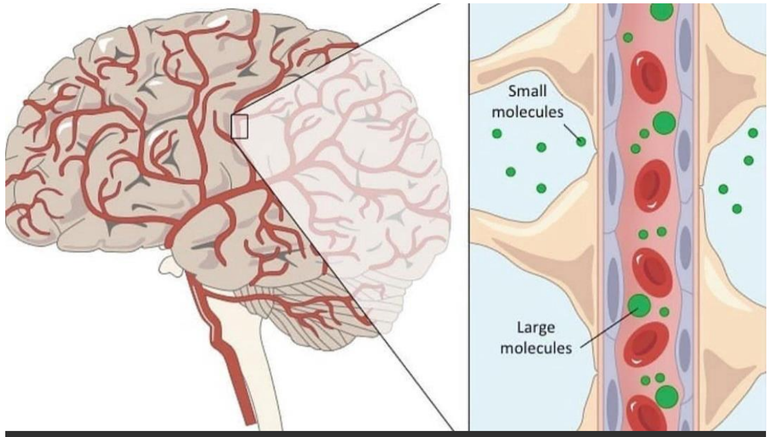


ChatGPT/ Bloom
(SINICA, HHRI, MTK)

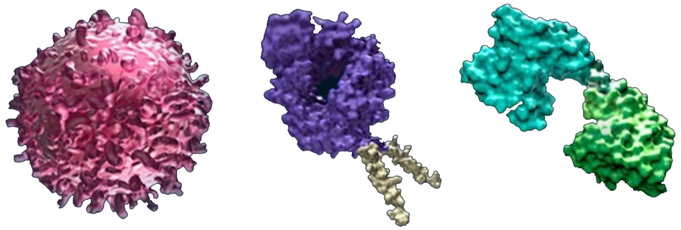


LLM/ Foundation Models Forums
(NTU, SINICA, NYCU, NTHU, HHRI, TWS)

ChatGPT is the “iPhone moment” for LLM/Generative AI”



Blood Brain Barrier with MegaMolBART
(CMUH, NCHC/TWS , NVAITC Joint Lab)



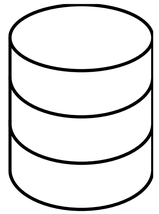
Anti-body – Dr Liu
(NHRI, NTU, NVAITC Joint Lab)

Gartner industry analyst predicts 30%+ of new drugs in 2025 will be created by LLM

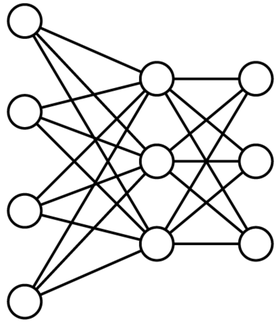
Reference:
1. iPhone moment: https://fortune.com/2023/03/01/artificial-intelligence-ai-chatgpt-iphone-moment-bank-of-america/
2. Gartner industry analyst: https://www.gartner.com/en/articles/beyond-chatgpt-the-future-of-generative-ai-for-enterprises
3. 大型語言模型線上技術論壇: https://tws.twcc.ai/2023/01/18/llm/

What is Large Language Model?

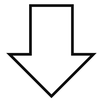
- **ChatGPT said**: “A large language model (LLM) is a language model that consists of a neural network with many parameters (usually **billions of weights** or more), and is trained on **large amounts of unlabeled text** using self-supervised learning. LLMs can recognize, summarize, translate, predict and generate text and other content.”



Large amounts of text data or sequence data



Billions of weights
GPT-3: 175 B
BLOOM: 176 B



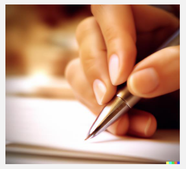
Content Generation



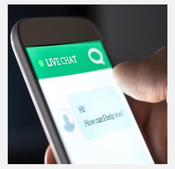
Summarization



Translation



Chatbots



Virtual Avatars

NVIDIA.

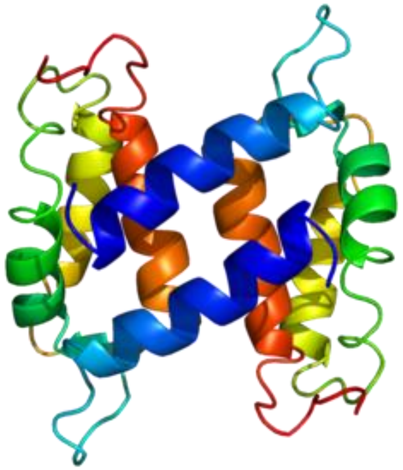
Relationship between Biology & LANGUAGE MODELS

Natural Language

Drug development is a cross-disciplinary endeavor

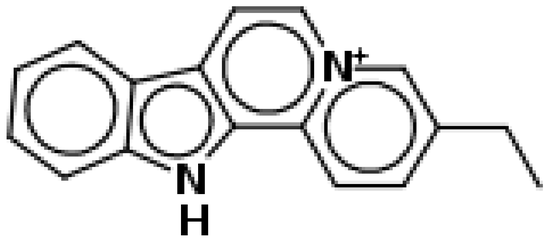
https://catalog.ngc.nvidia.com/orgs/nvidia/collections/claradiscovery

Language-like Series



```
ATIKVTKANTLFKIAAY
PMMVAKKRIILRAMKRI
ILRIKVTKANTLFKIIL
RIKVTNTLFKIILRIKV
TNTMVAKKRITNKLISL
ALALIRMKTLILRIKVT
KAVTKANTLFKIILRIK
VGGKANTLFKIILRIKV
```

FASTA Sequence



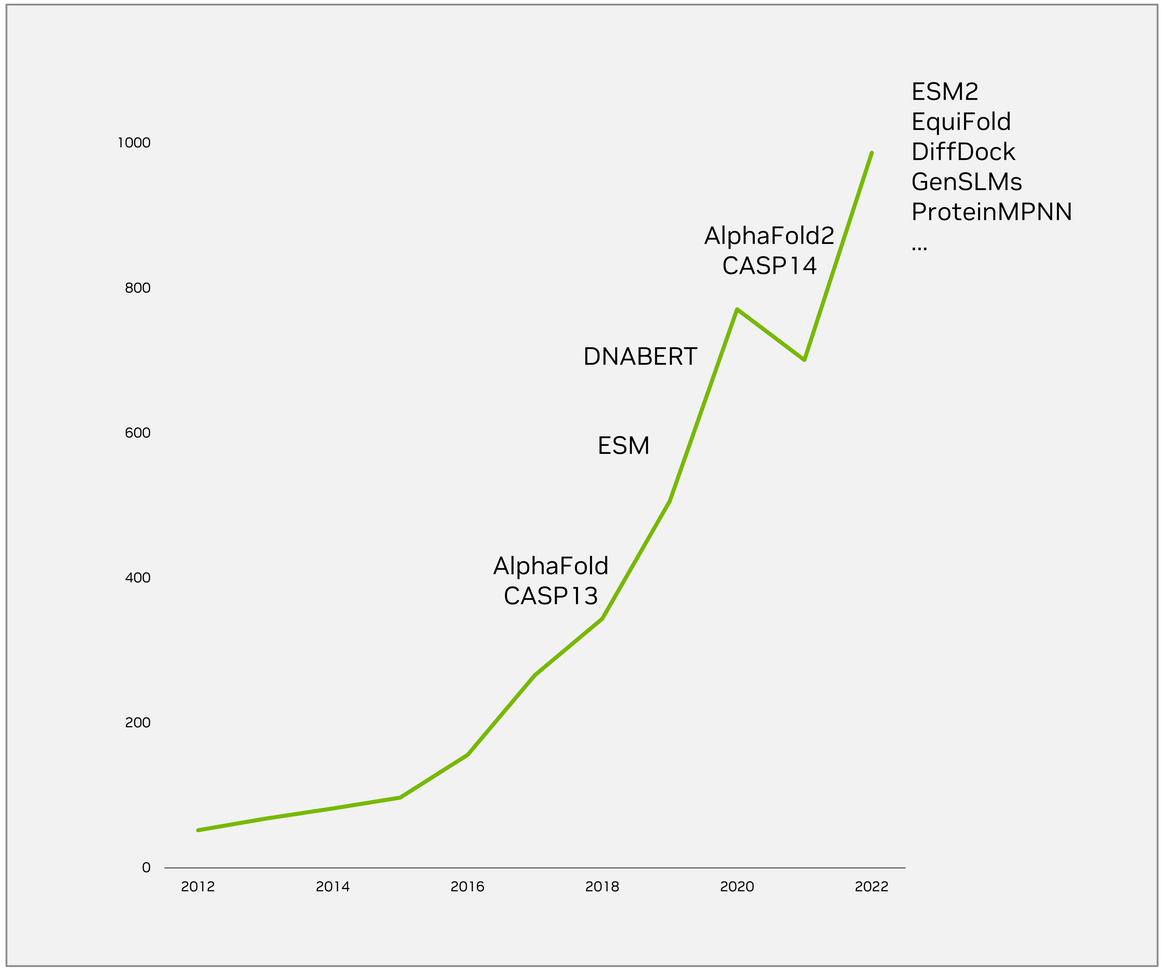

```
CCc(c1)ccc2[n+]1ccc3c2[nH]c4c3cccc4
CCc1c[n+]2ccc3c4ccccc4[nH]c3c2cc1
```

SMILES

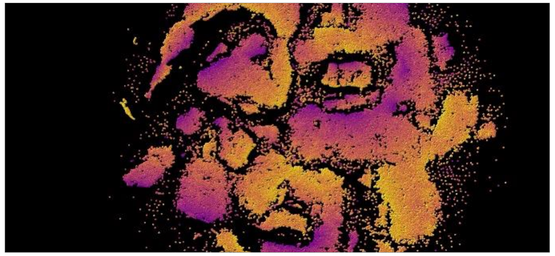
NVIDIA.

Generative AI is Exploding in Drug Discovery

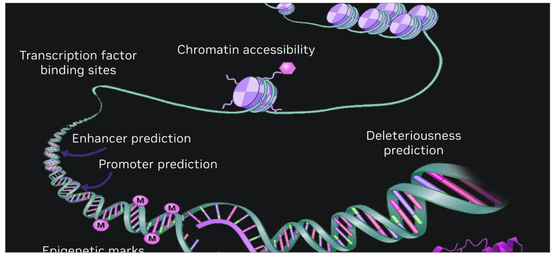
AI Published Papers



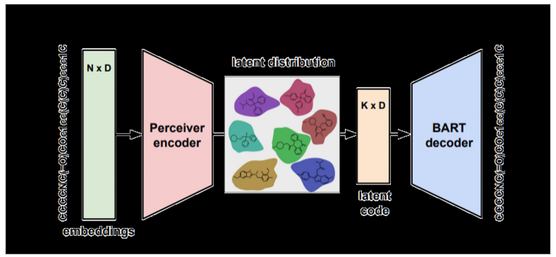
NVIDIA Joint Research



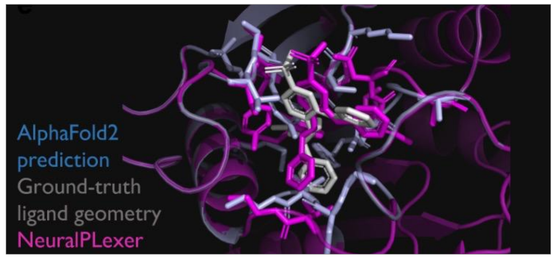
GensLMs
Genome-scale language models reveal SARS-CoV-2 Evolutionary Dynamics



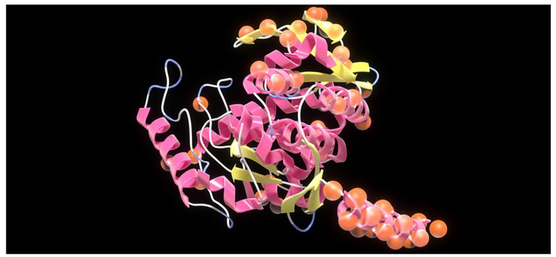
Nucleotide Transformer
Building and Evaluating Robust Foundation Models for Human Genomics



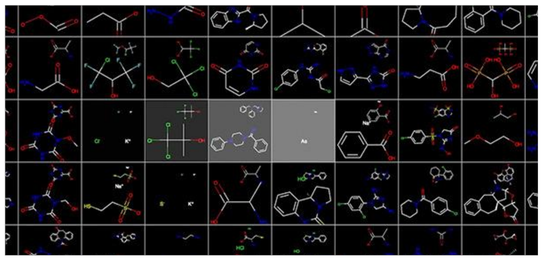
MolMIM
Improving Small Molecule Generation using Mutual Information Machine



NeuralPlexer
Dynamic-Backbone Protein-Ligand Structure Prediction with Multiscale Generative Diffusion Models



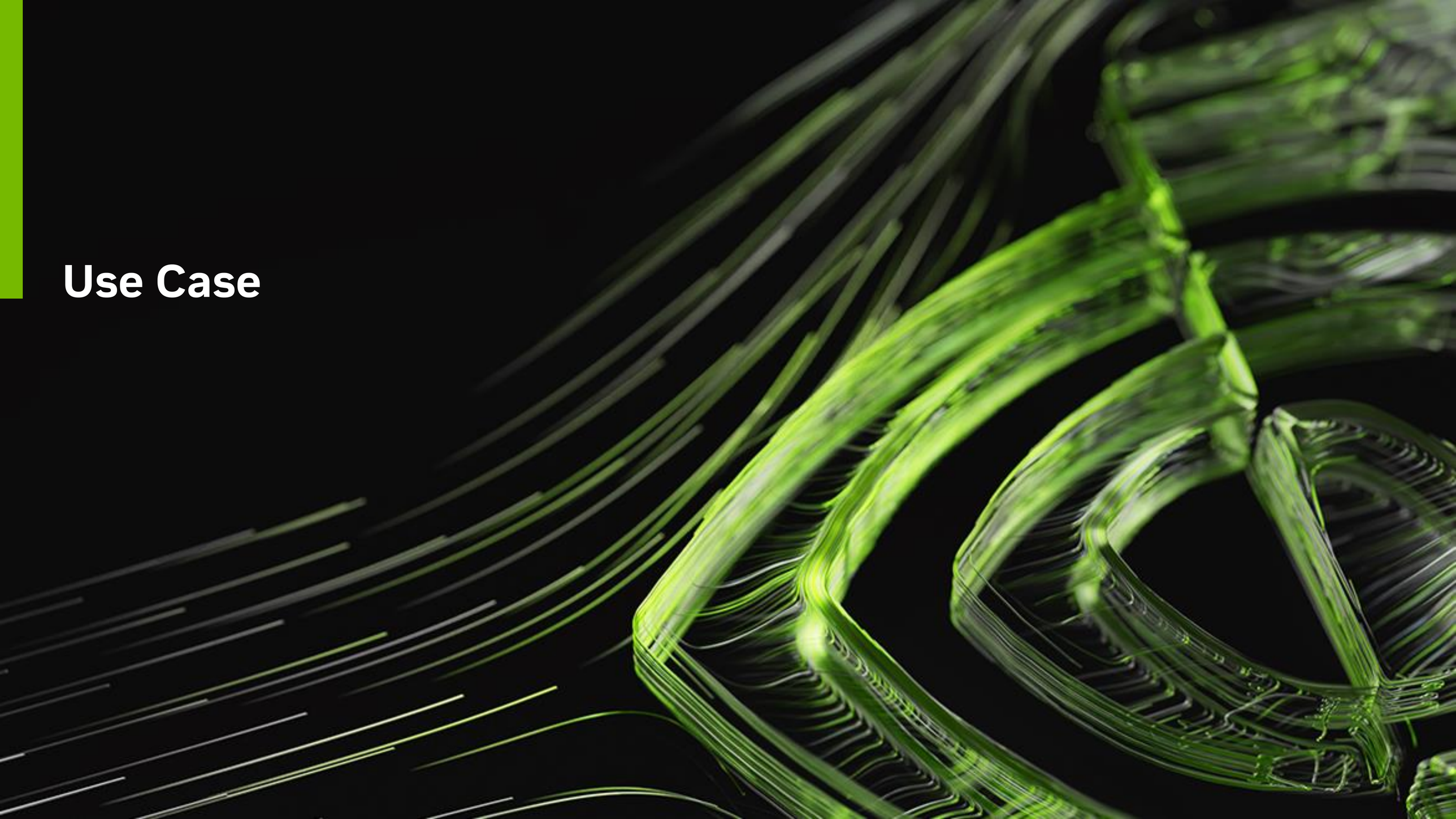
ProT-VAE
Protein Transformer Variational AutoEncoder for Functional Protein Design



MegaMolBART
Generally Applicable Chemical AI Models with Large-Scale Pretrained Transformers

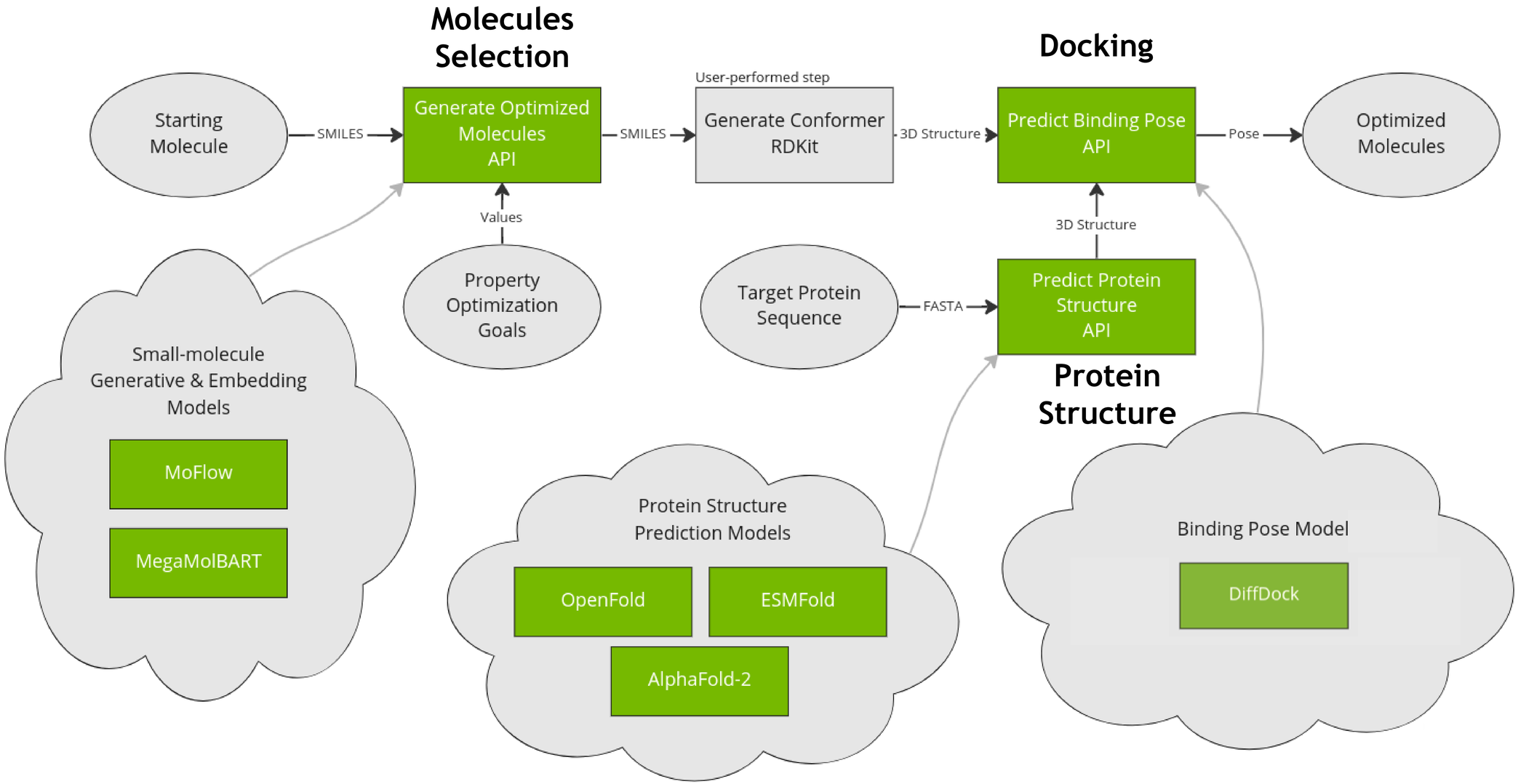
Source: arXiv.org Q-bio: AI, ML, DL, NN

Use Case

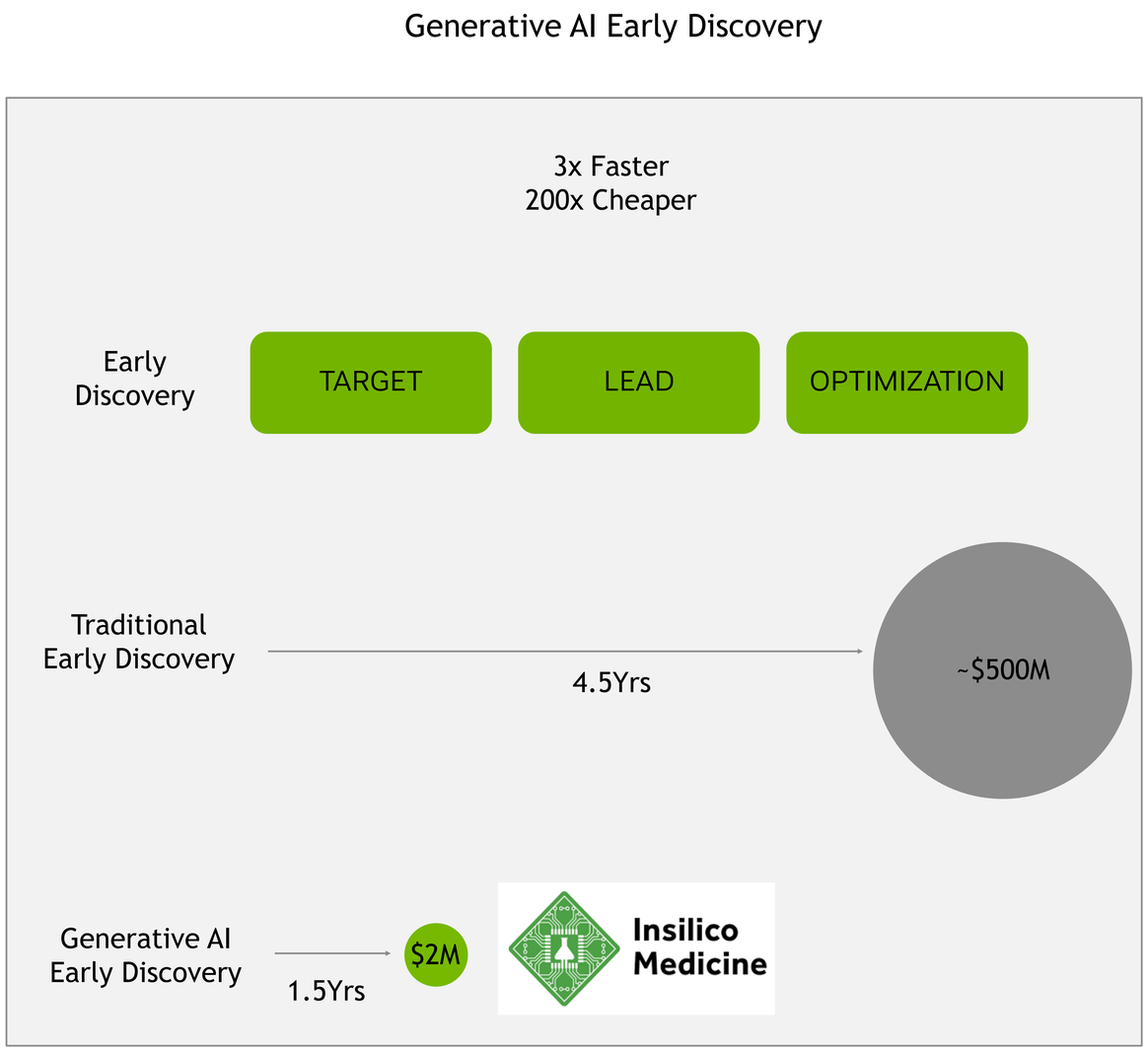


Models for AI-Powered Virtual Screening

DiffDock | Pose Prediction Significantly Exceeding State-of-the-Art Methods



Insilico Medicine: Generative AI is Accelerating Early Drug Discovery



EurekAlert! | AAAS

HOME NEWS RELEASES MULTIMEDIA MEETINGS

NEWS RELEASE 8-FEB-2023

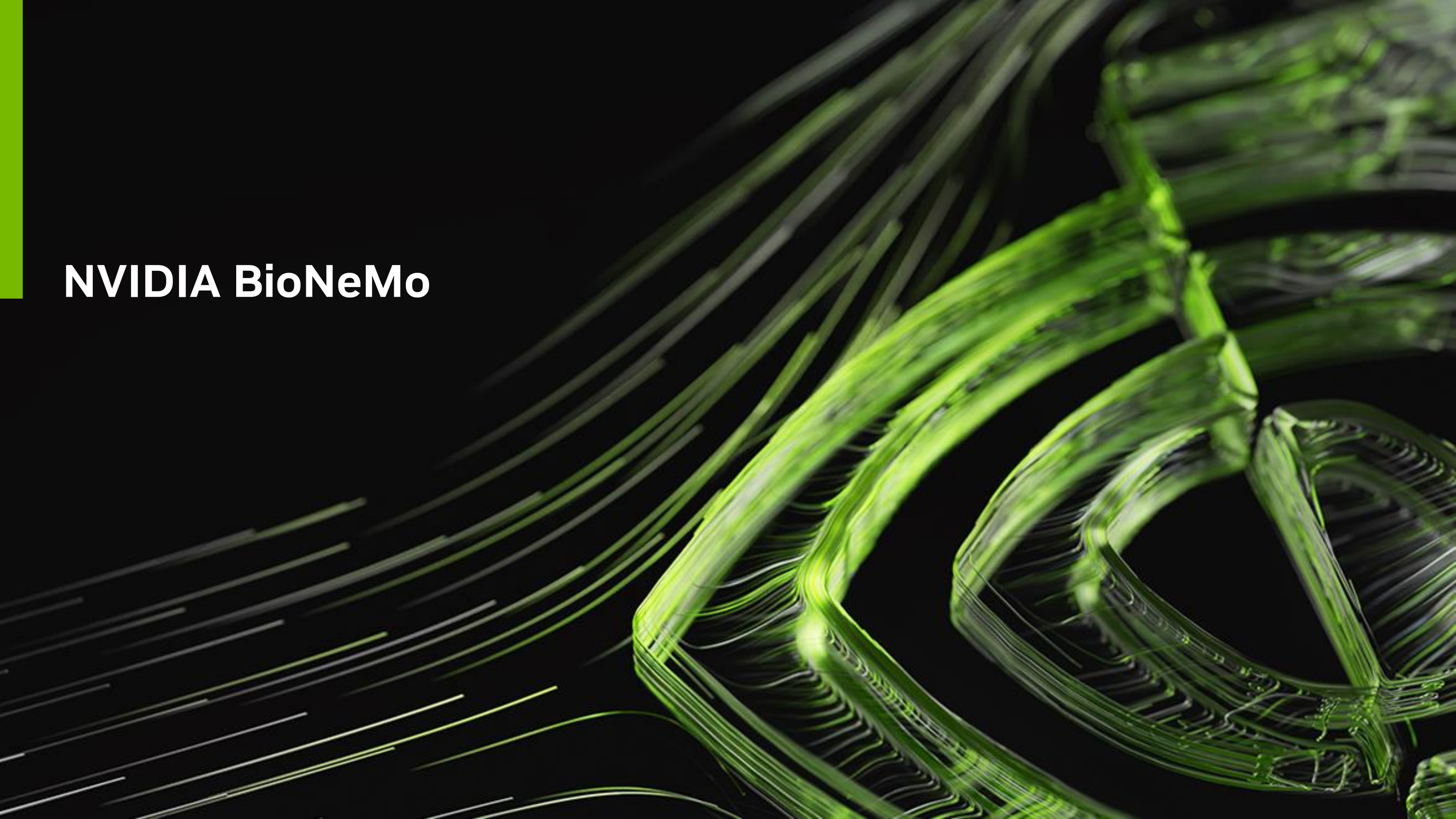
Insilico Medicine receives FDA Orphan Drug Designation for generative AI discovered and designed drug for idiopathic pulmonary fibrosis

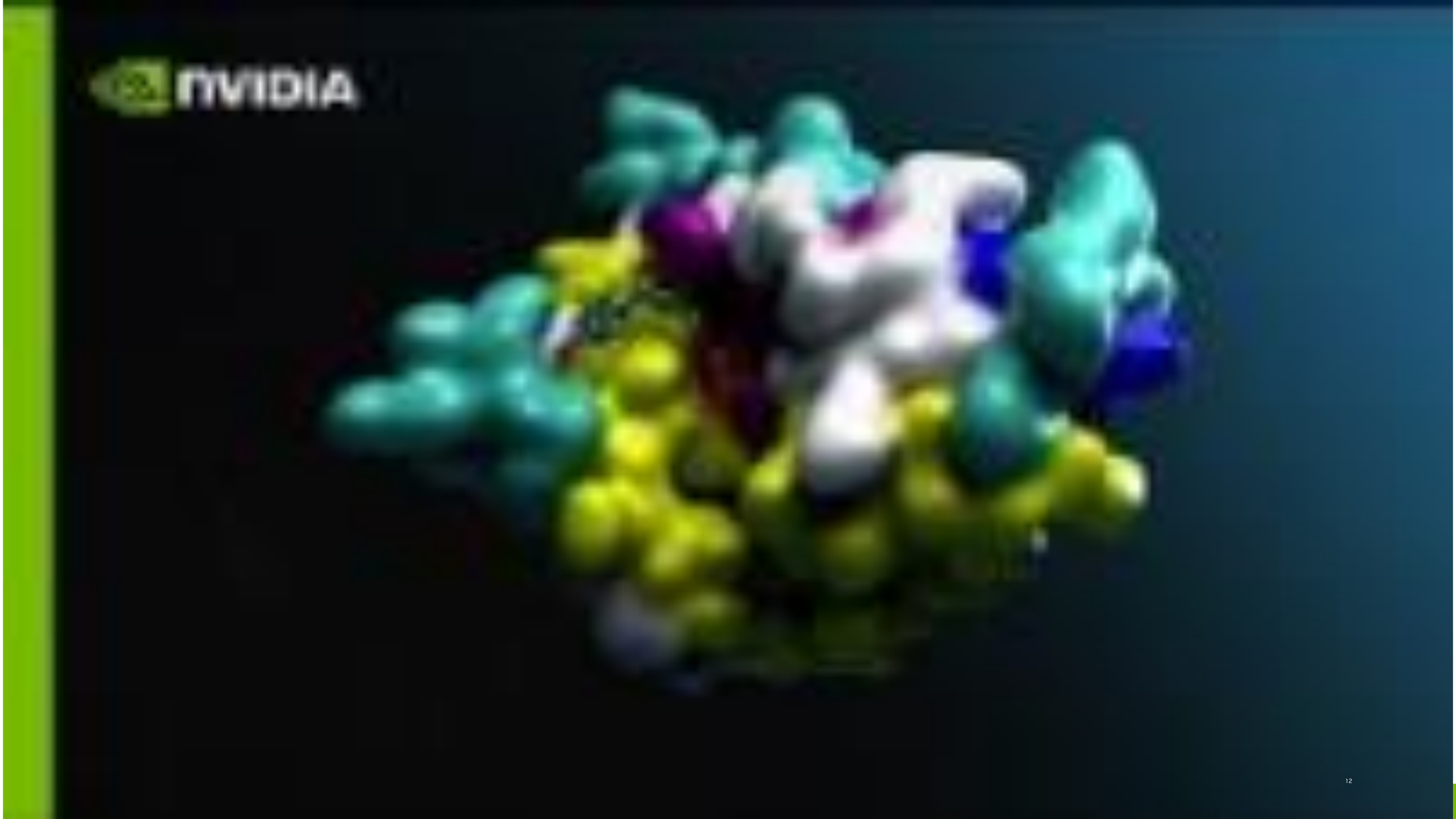
Business Announcement

INSILICO MEDICINE

https://www.eurekalert.org/news-releases/979045

NVIDIA BioNeMo

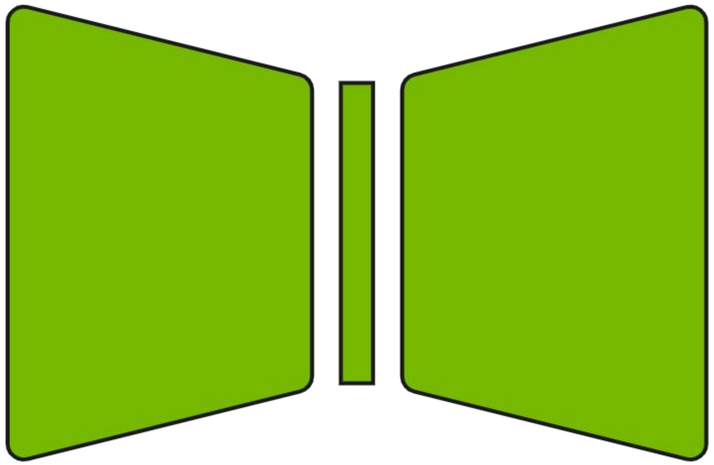


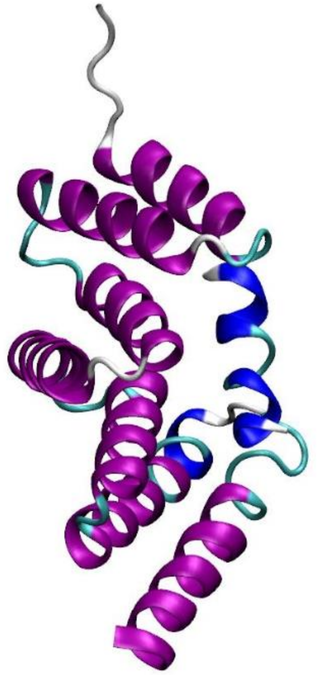

NVIDIA

BioNeMo – Protein Structure Prediction

AlphaFold2, OpenFold, ESMFold | Fold Proteins with BYO MSA or MSA-Free

MHHHHHHSSGLVPRGSGMKETAAAKFERQH
MDSPDLGTDDDDKAMADIGSENLYFQSMSK
IFVNPSAIRAGMADLEMAEETVDLINRNIE
DNQAHLQGEPIEVDSLPEDIENLYFQGMES
DKIVFKVNNQLVSVKPEVIVDQYEYKYPAI
QDHTKPSITLGKAPDLNKAYKSILSGMNAA
KLDPDDVCSYLAAAMELFEGVCPEDWTSYG
IMIARKGDKITPATLVDIKRTDIEGNWALT
GGQDLTRDPTVAEHASLVGLLLSLYRLSKI
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GFIKQIN

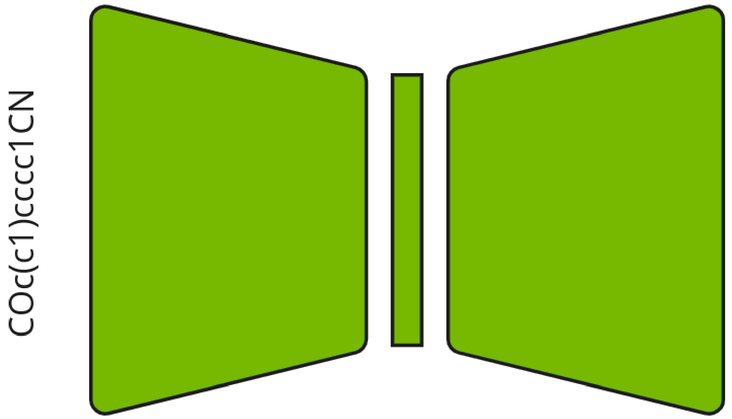


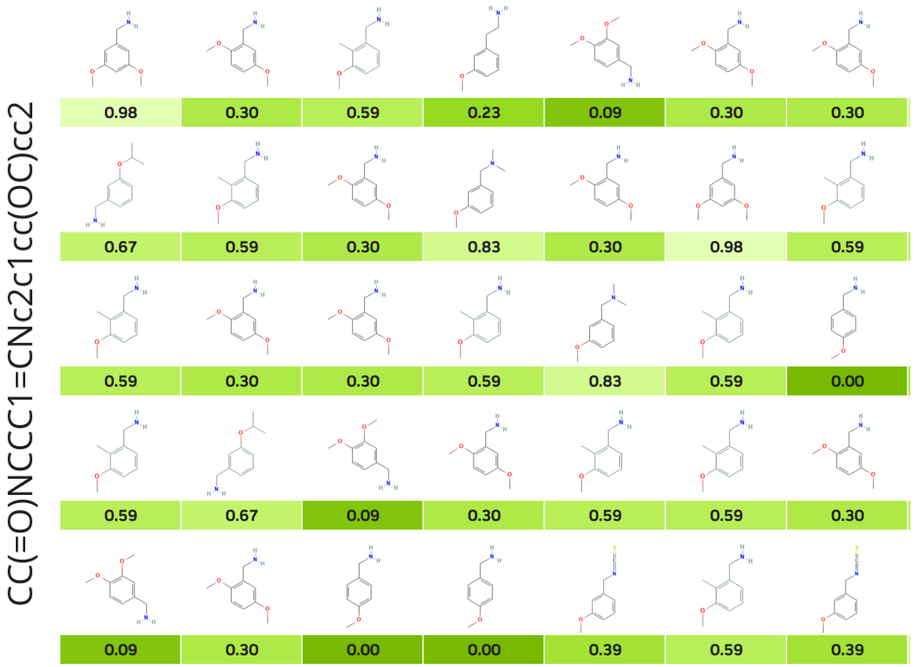


AlphaFold-2: DeepMind; Jumper et al.
RoseTTAFold: UW IPD; Baker et al.
OpenFold: OpenFold Consortium; AlQuraishi et al.
ESMFold: MetaAI; Lin et al.

BioNeMo – Molecule Generation

MoFlow & MegaMolBART | Generate Molecules with Optimized Properties

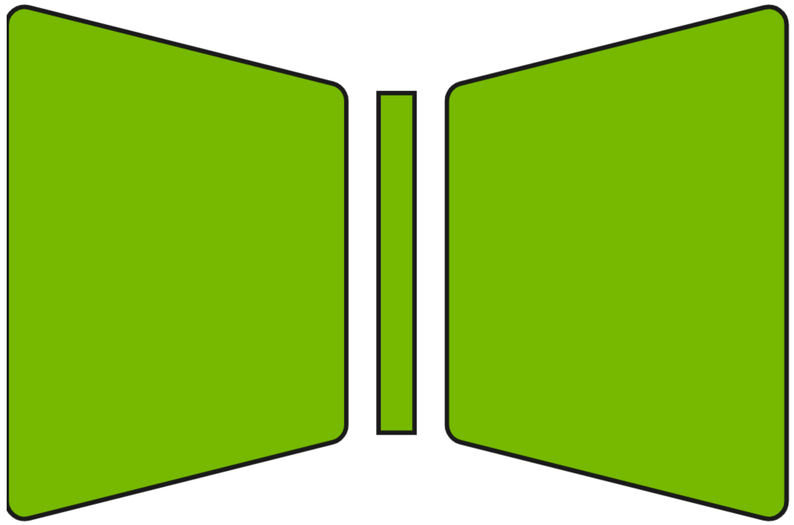




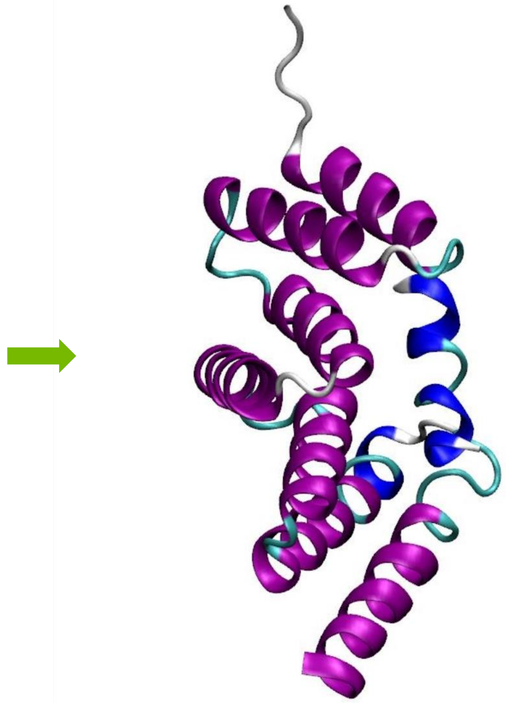
MoFlow: Chengxi Zang and Fei Wang
MegaMolBART: NVIDIA & AstraZeneca; Bjerrum, Irvin, Engkvist, et al.

Protein Sequence Generation

ProtGPT2



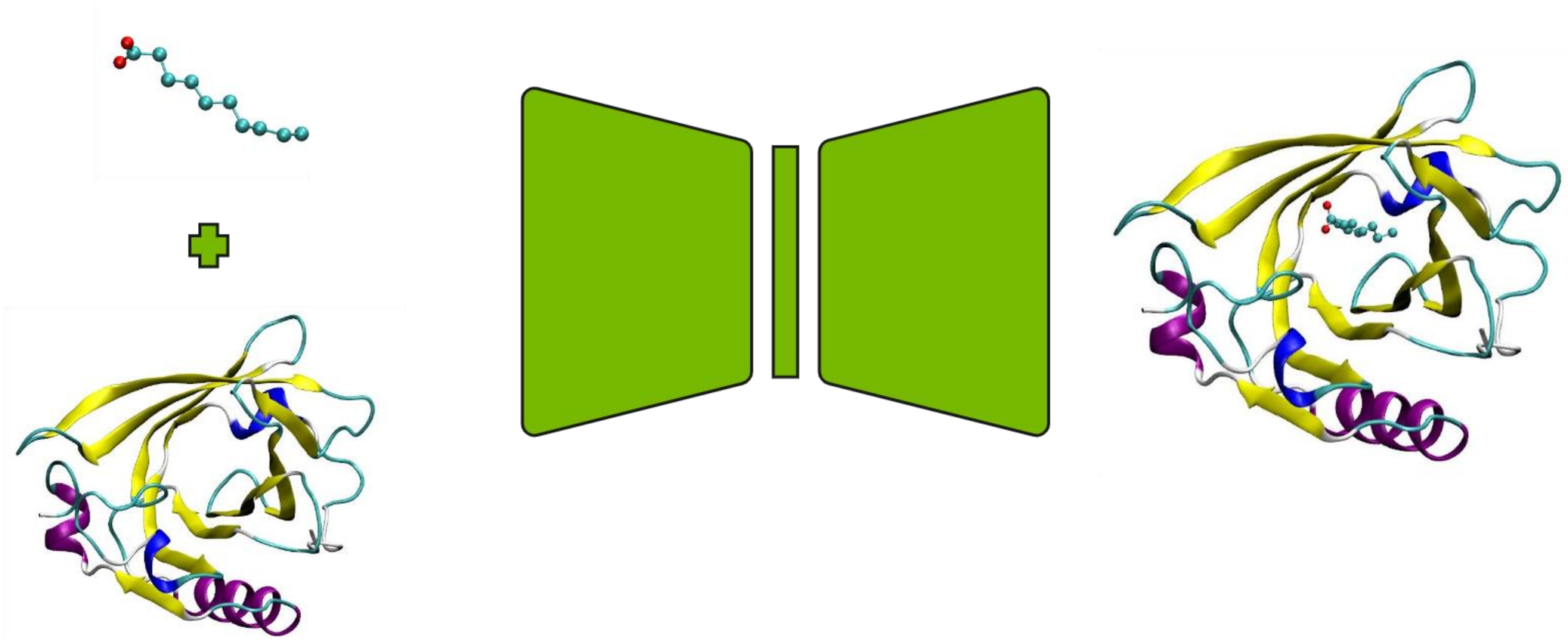
MHHHHHSSGLVPRGSGMKETAAAKFERQHMDSPDL
GTDDDDKAMADIGSENLYFQSMSKIFVNPSAIRAGM
ADLEMAEETVDLINRNIEDNQAHLQGEPIEVDSLPE
DIENLYFQGMESDKIVFKVNNQLVSVKPEVIVD...



Noelia Ferruz, Steffen Schmidt & Birte Höcker

BioNeMo – Docking Pose Prediction

DiffDock | Pose Prediction Significantly Exceeding State-of-the-Art Methods

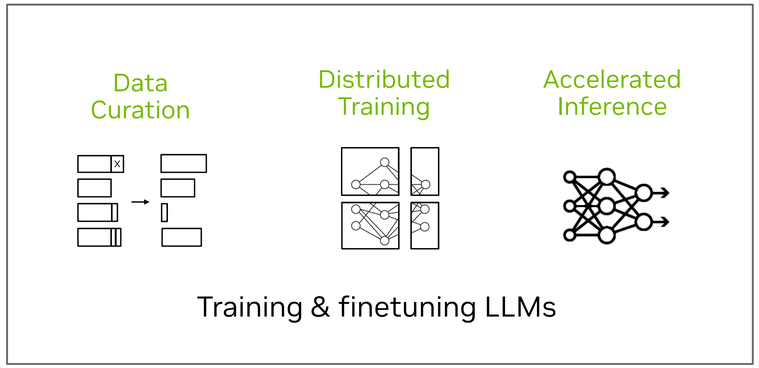


38% top-1 success rate (RMSD<2A) on PDB-Bind

Gabriele Corso, Hannes Stark, Bowen Jing, Regina Barzilay & Tommi J

NVIDIA BioNeMo Framework / Service

BioNeMo Training



AI Models in BioNeMo

- MegaMolBART is LLM for state-of-the-art **small molecule generation**
- MoFlow is for **generative chemistry**
- ESM1nv is LLM for **protein embeddings** for downstream tasks
- ESM2 is LLM for **protein embeddings** for downstream tasks
- ProtGPT2 is for **protein generation**
- DiffDock is for molecular **docking pose estimation**

Domain Specific

Drug discovery researchers need an LLM framework that speaks the language of biology and chemistry.

Optimized For Scale

LLMs are massive requiring model parallelism and supercomputing resources to train.

Pre-Trained Models

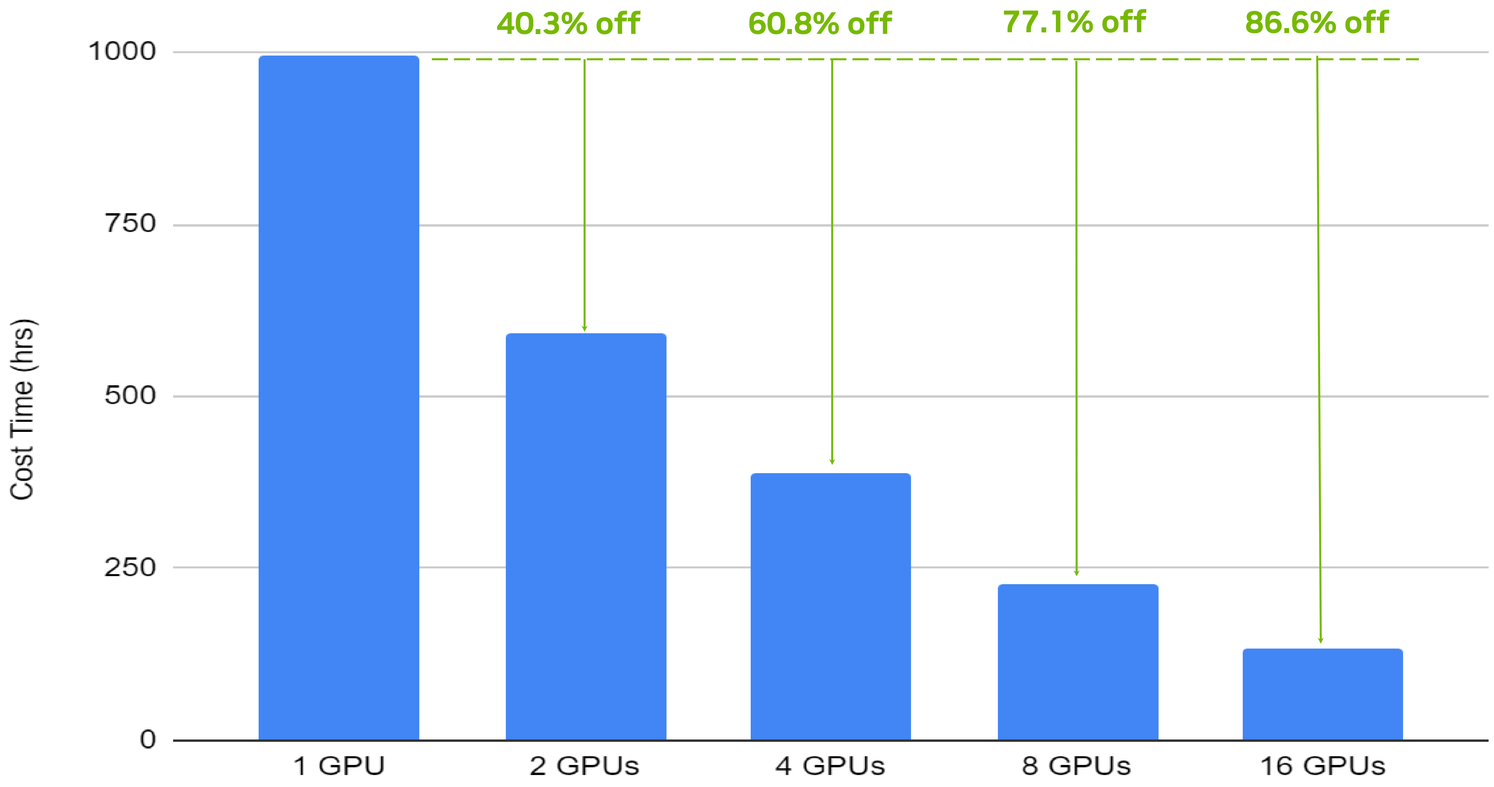
Training LLMs takes enormous compute resources and is very time consuming. Pre-trained models are ready for production.

Cloud Native

Use pre-trained models, create powerful customized workflows and deploy in the cloud.

Language Model Multi-GPU Benchmarking

Model = 384m parameters, Data = 6B protein sequences, Machine: DGX A100



	1 GPU	2 GPUs	4 GPUs	8 GPUs	16 GPUs (2 nodes)
Cost Time	994 hrs	593 hrs	390 hrs	228 hrs	133 hrs

*With 160 A100s:
- GPT-3 5B: ~ 5 days
- T5 3B: ~ 11 days

*With 160 H100s:
- GPT-3 5B: ~ 1.x days
- T5 3B: ~ 3.x days

MONAI Generative AI

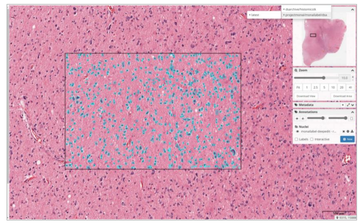


MONAI

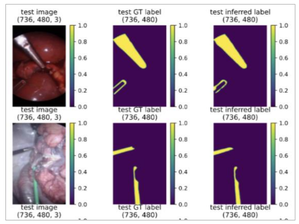
Industry Standard Medical AI Framework



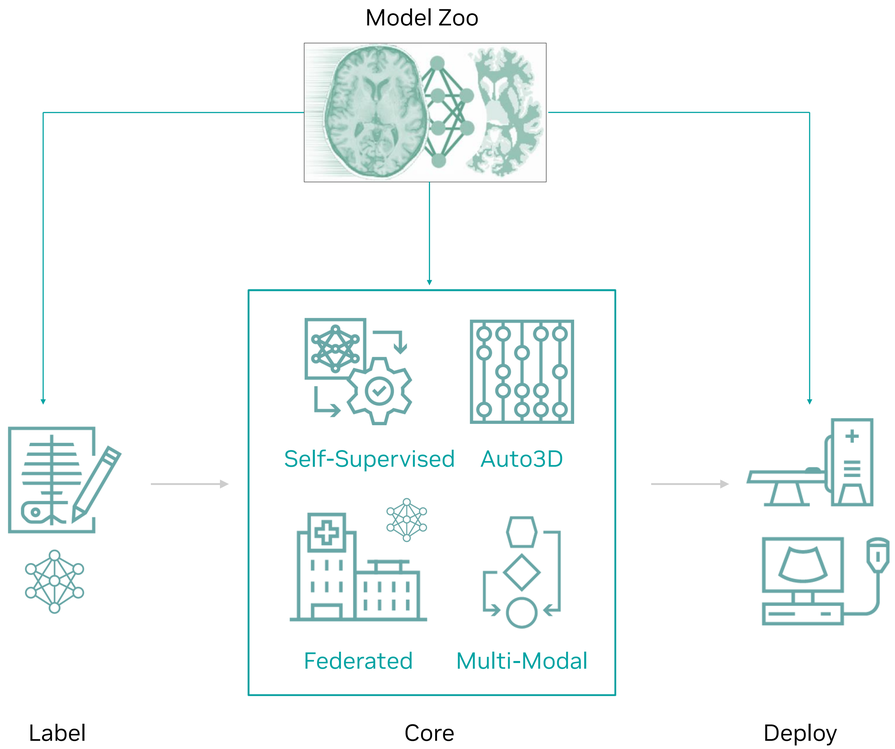
Radiology



Pathology

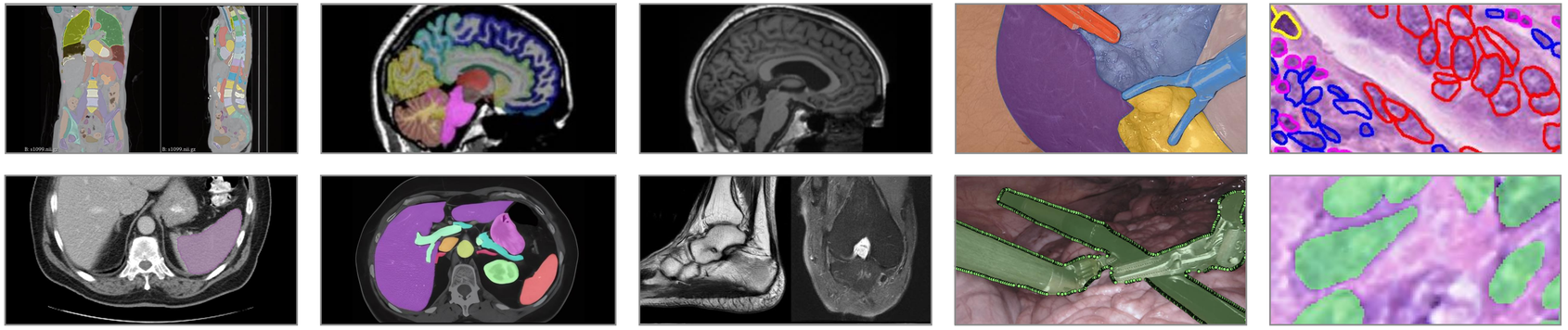


Endoscopy



MONAI Model Zoo

Plug Into Any MONAI Workflow from Labeling to Training to Deployment

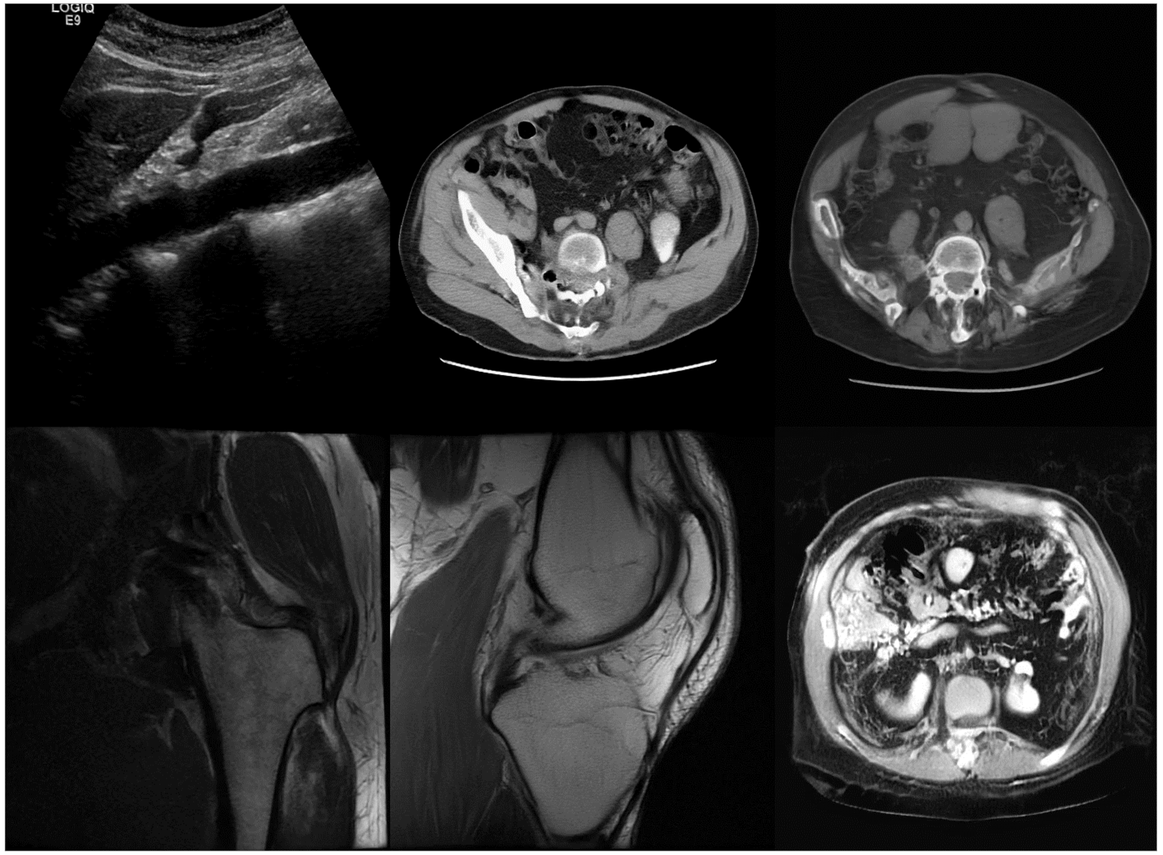


CT MRI 3D SYNTHETIC DATA SURGICAL VIDEO PATHOLOGY

NVIDIA AI

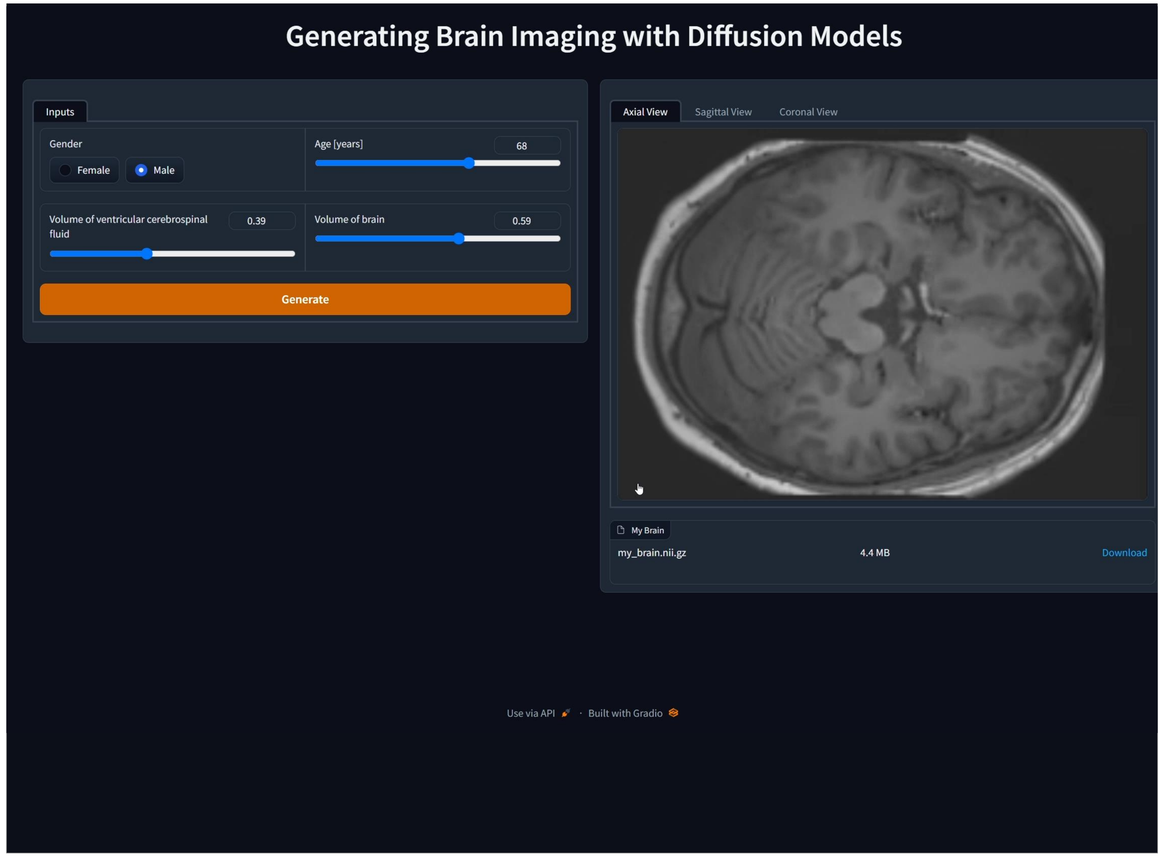
MONAI Generative AI

Create Infinite Medical Imaging Data with Generative AI



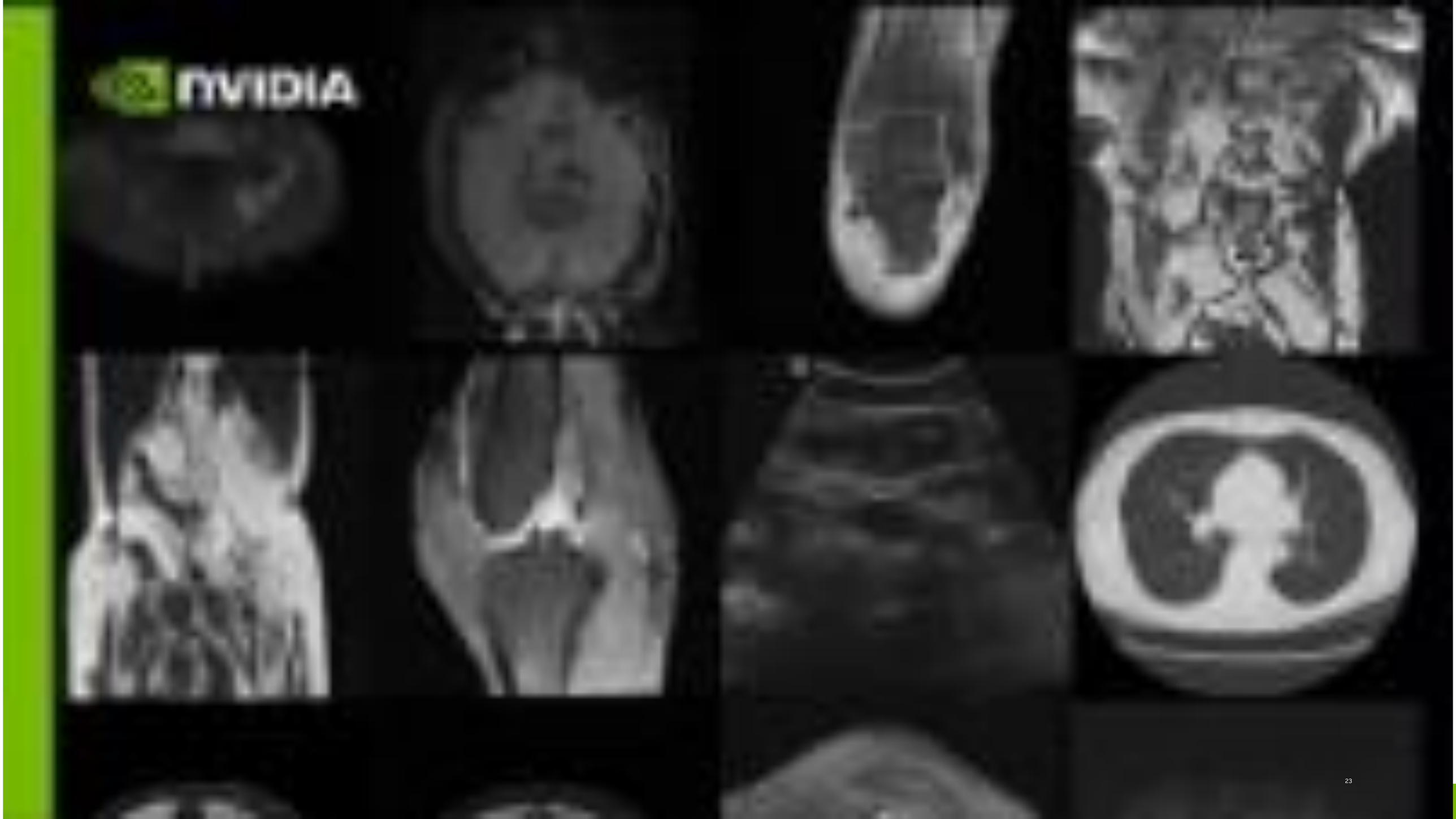
RADIMAGEGAN

3 modalities CT, MRI, US, 14 anatomies, and 165 pathologic labels
NVIDIA | Mt. Sinai | East River Imaging
Available NGC Catalog



Latent Diffusion Model

High-resolution brain images | Targeted datasets for brain volume, age, disease
NVIDIA | Kings College London
Available NGC Catalog | MONAI Model Zoo


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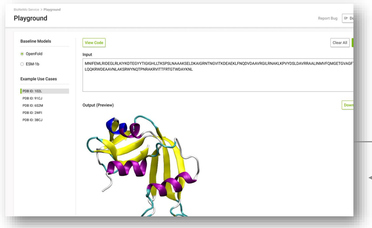
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https://www.nvidia.com/en-us/startups


NVIDIA
Thank You

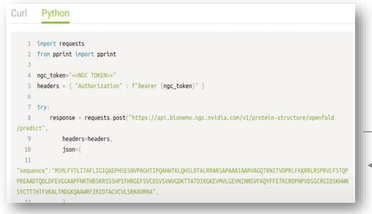
BioNeMo

For Generative AI in Biology - Supporting 9 SOTA Small Molecule & Protein AI Models at HPC Scale

BioNeMo



WEB



API

Drug Discovery Applications

Protein Structure Prediction	Biomolecule Representation
Biomolecule Generation	Molecular Docking

BioNeMo Models

AlphaFold2	OpenFold	ESMFold	ESM1	ESM2
ProtGPT2	MegaMolBART	MoFlow	DiffDock	

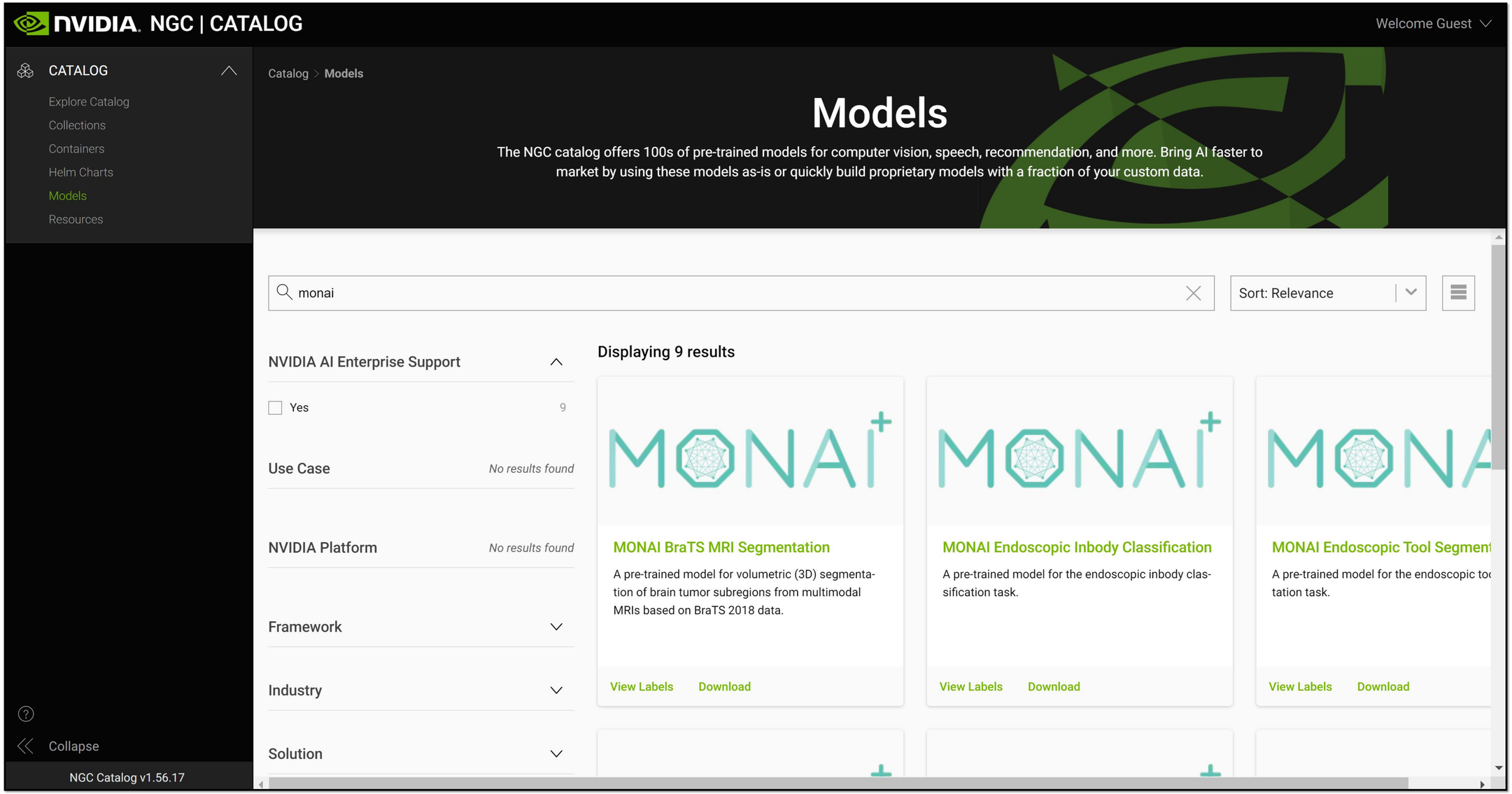
NVIDIA DGX Cloud



Training, fine-tuning, inferencing and deploying

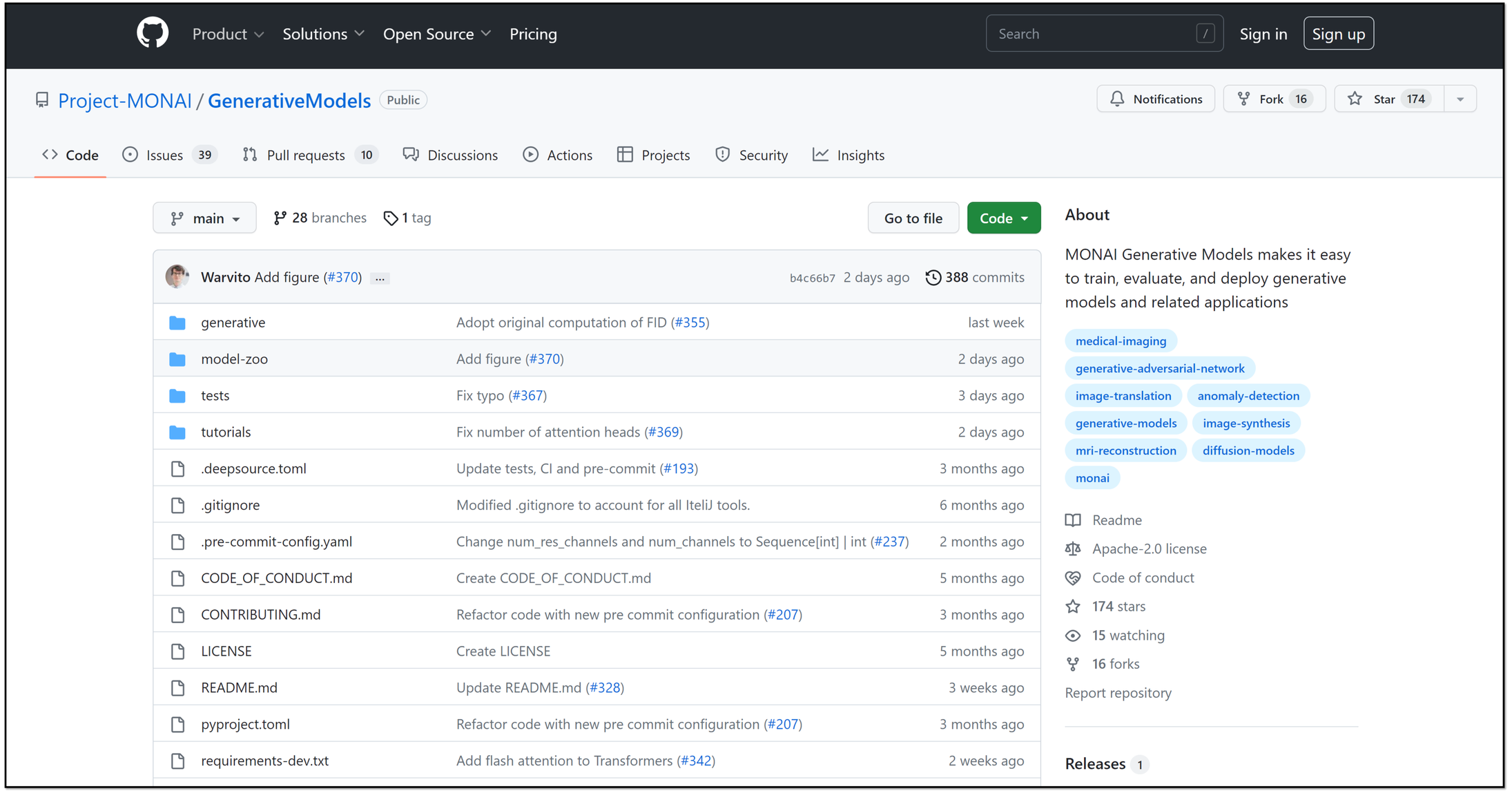
NVIDIA GPU Cloud

Free to register and download



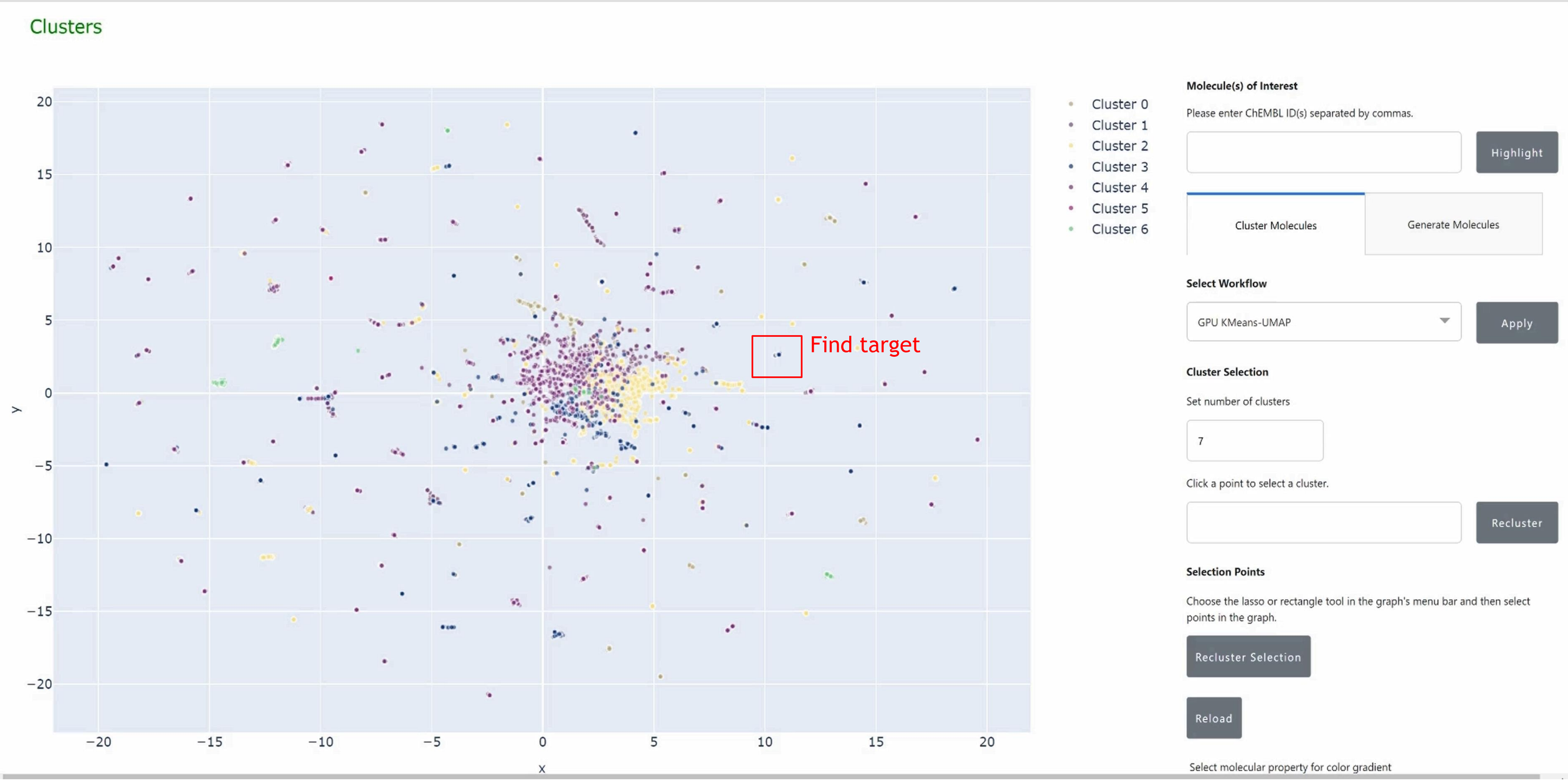
MONAI Gen AI Official GitHub

https://github.com/Project-MONAI/GenerativeModels



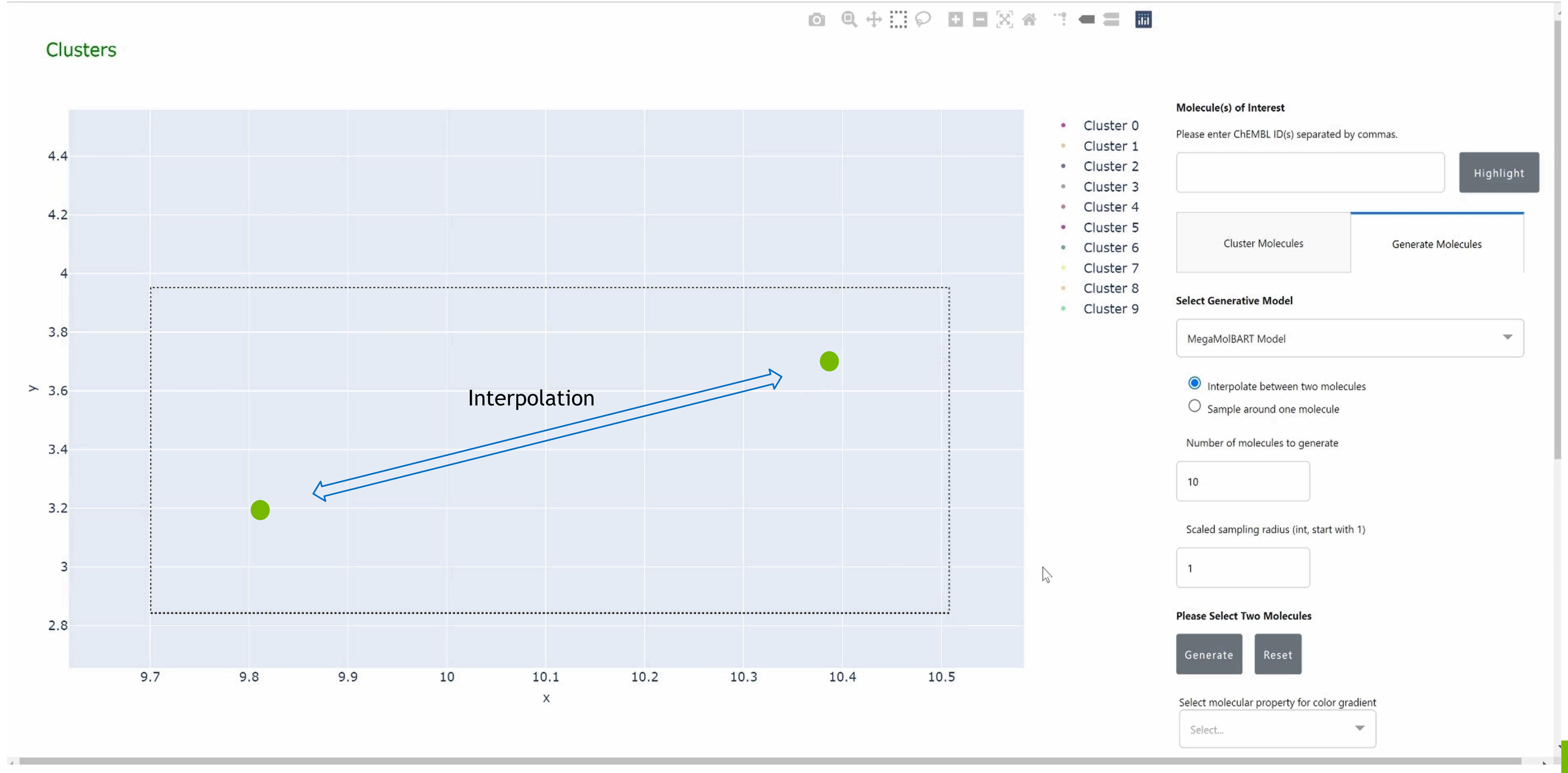
Molecules Discovery Demo - MegaMolBART

https://github.com/NVIDIA/cheminformatics



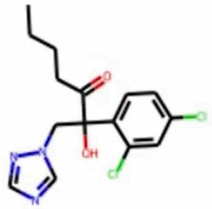
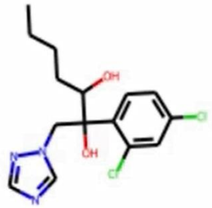
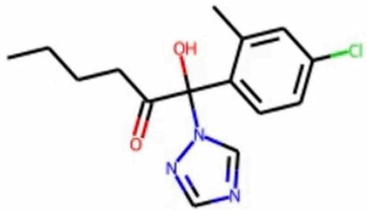
Molecules Discovery Demo - MegaMolBART

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Molecules Discovery Demo - MegaMolBART

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SMILES	Generated	Chemical Structure	Molecular Weight	LogP	H-Bond Donors	H-Bond Acceptors	Rotatable Bonds	QED
CCCCC(=O)C(O)(Cn1cncn1)c1ccc(Cl)cc1Cl	False		342.23	3.23	1	5	7	0.8388
CCCCC(O)C(O)(Cn1cncn1)c1ccc(Cl)cc1Cl	True		344.24	3.02	2	5	7	0.81
CCCC(=O)C(O)(Cn1cncn1)c1ccc(Cl)cc1Cl	True		328.2	2.84	1	5	6	0.8853
CCCCC(=O)C(O)(n1cncn1)c1ccc(Cl)cc1C	True		307.78	2.69	1	5	6	0.8907